Integrated Programme – JEE MAIN Revision Sheet

NLM & Friction

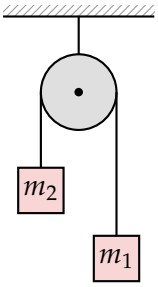
Level 1 Sheet

Time to be taken: 60 Min Max

Q1. Two blocks of masses m_1 and m_2 are attached at the ends of an inextensible string which passes over a smooth massless pulley. If $m_1 > m_2$, find:

(i) the acceleration of each block

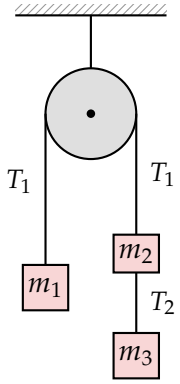
(ii) the tension in the string.



Q2. A light unstretchable string passing over a smooth light pulley connects two blocks of masses m_1 and m_2. If the acceleration of the system is $\frac{g}{8}$, then the ratio of the masses $\frac{m_2}{m_1}$ is :

(A) 8 : 1 (B) 5 : 3 (C) 4 : 3 (D) 9 : 7

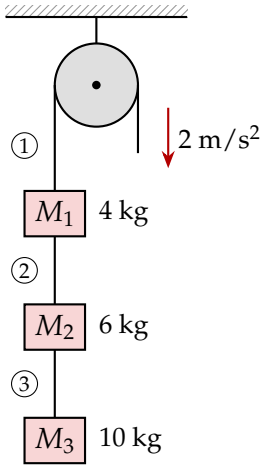
Q3. Three masses $m_1 = 4$ kg, $m_2 = 4$ kg and $m_3 = 6$ kg are suspended from a fixed smooth frictionless pulley as shown in the figure. The value of T_1/T_2 is _____.
(take $g = 10\ \text{m/s}^2$)



(Pulley Concept and Taking System Concept)

Note – Better to revise pulley with some moment of inertia questions here from Rotational Motion.

Q4. Three blocks M_1, M_2, M_3 having masses 4 kg, 6 kg and 10 kg respectively are hanging from a smooth pulley using rope 1, 2 and 3 as shown in figure. The tension in the rope 1, T_1 when they are moving upward with acceleration of 2 m s^{-2} is _____ N (if $g = 10$ m/s^2).



Q5. At $t = 0$, a body of mass 100 g starts moving under the influence of a force $(5\hat{i} + 10\hat{j})$ N. After 2 s its position is $(2x\hat{i} + 5y\hat{j})$ m. The ratio $x : y$ is _____.

(A) 1 : 2 (B) 2 : 5 (C) 5 : 2 (D) 5 : 4

(Basic 2-D Concept, X and Y independent of each other)

Q6. A 4 kg mass moves under the influence of a force $\vec{F} = \left(4t^3\hat{i} - 3t\hat{j}\right)$ N where t is the time in second. If mass starts from origin at $t = 0$, the velocity and position after $t = 2$ s will be:

(A) $\vec{v} = 4\hat{i} - \frac{3}{2}\hat{j}$ $\vec{r} = \frac{6}{5}\hat{i} - \hat{j}$ (B) $\vec{v} = 3\hat{i} + \frac{3}{2}\hat{j}$ $\vec{r} = \frac{6}{5}\hat{i} + \hat{j}$
(C) $\vec{v} = 4\hat{i} - \frac{3}{2}\hat{j}$ $\vec{r} = \frac{8}{5}\hat{i} - \hat{j}$ (D) $\vec{v} = 4\hat{i} + \frac{5}{2}\hat{j}$ $\vec{r} = \frac{8}{5}\hat{i} + 2\hat{j}$

(Basic 2-D Concept, X and Y independent of each other)

Q7. An object with mass 500 g moves along x-axis with speed $v = 4\sqrt{x}$ m/s. The force acting on the object is :

(A) 8 N (B) 5 N (C) 6 N (D) 4 N

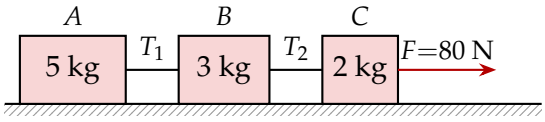
(Basic Calculus of x, v, t and a)

Q8. A cricket player catches a ball of mass 120 g moving with 25 m s^{-1} speed. If the catching process is completed in 0.1 s then the magnitude of force exerted by the ball on the hand of player will be (in SI unit):

(A) 24 (B) 12 (C) 25 (D) 30

(Basic 2nd Law application)

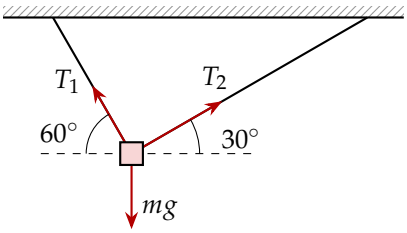
Q9. Three blocks A, B and C are pulled on a horizontal smooth surface by a force of 80 N as shown in figure. The tensions T_1 and T_2 in the string are respectively:



(A) 40 N, 64 N
(B) 60 N, 80 N
(C) 88 N, 96 N
(D) 80 N, 100 N

(Multiple Block and String Question)

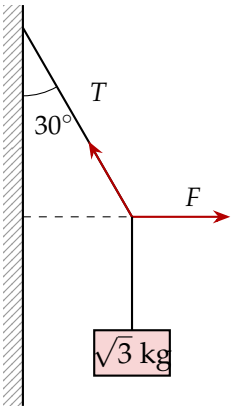
Q10. A body of mass 1 kg is suspended with the help of two strings making angles as shown in figure. Magnitude of tensions T_1 and T_2, respectively, are (in N) :



(A) 5, $5\sqrt{3}$
(B) $5\sqrt{3}$, 5
(C) $5\sqrt{3}$, $5\sqrt{3}$
(D) 5, 5

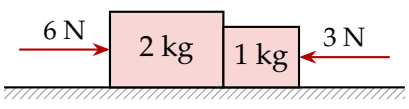
(Equilibrium Concept)

Q11. A block of $\sqrt{3}$ kg is attached to a string whose other end is attached to the wall. An unknown force F is applied so that the string makes an angle of 30° with the wall. The tension T is : (Given $g = 10\ \text{m s}^{-2}$)



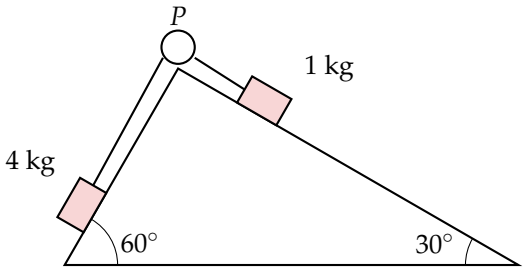
(A) 20 N
(B) 25 N
(C) 10 N
(D) 15 N

Q12. Two forces of 6 N and 3 N are acting on the two blocks of 2 kg and 1 kg kept on frictionless floor. What is the force exerted on 2 kg block by 1 kg block ?



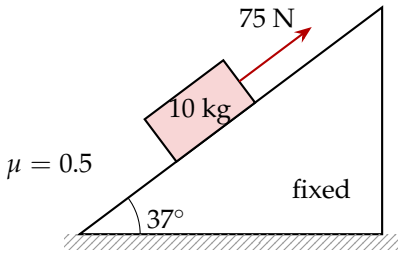
(A) 1 N
(B) 2 N
(C) 4 N
(D) 5 N

Q13. As per given figure, a weightless pulley P is attached on a double inclined frictionless surface. The tension in the string (massless) will be (if $g = 10\ \text{m s}^{-2}$)



(A) $\left(4\sqrt{3}+1\right)$ N (B) $4\left(\sqrt{3}+1\right)$ N (C) $4\left(\sqrt{3}-1\right)$ N (D) $\left(4\sqrt{3}-1\right)$ N

Q14. Find out the acceleration of the block. If the block is initially at rest.



Note – Prepare Graphs very well, prepare all terminal velocities cases graph in syllabus. Always solve and find equations and then choose the correct answer (graph).

★ **KEEP GOING** ★

Every JEE topper was once a beginner who refused to quit. Draw the free-body diagram, trust the process, and solve one more problem today than you did yesterday — your effort today is your rank tomorrow.